Simulation Report

Experiment 3 - Pipelined AES-256 Encryption Engine on the PYNQ-Z2

1. Setup

Item	Value
Simulator	Icarus Verilog 12 (iverilog -g2012, SystemVerilog)
Design under test	aes_top (RTL/*.sv)
Testbench	Testbench/aes_tb.sv
Waveform dump	Simulation/wave_dump.sv (records the main signals to aes_wave.vcd)
Clock	100 MHz (10 ns period), same as the board
Test vector	FIPS-197 appendix C.3 (the official AES-256 example)

2. What the testbench checks

The testbench drives the AXI4-Lite port of **aes_top** the same way the ARM processor does on the board, with the tasks *axi_write* and *axi_read*, plus helpers that write or read a whole 128-bit block and start the engine and wait for done.

#	Step	Expected result
1	Write the key 000102...1e1f to 0x50-0x6C, write 4 to CTRL	Key schedule ready bit (STATUS bit 2) sets
2	Write 00112233...eeff to 0x20-0x2C, write 1 to CTRL (encrypt)	Result 8ea2b7ca516745bfeafc49904b496089 (FIPS-197)
3	Write the ciphertext, write 2 to CTRL (decrypt)	The original plaintext comes back
4	Read the key registers 0x50-0x6C	All read as zero (the key is write-only)
5	Read the block counter 0x40	2 blocks

Any wrong value stops the simulation with *$fatal*; reaching the last line means every check passed.

3. Result

PASS - the ciphertext matches the FIPS-197 test vector, decryption recovers the plaintext, the key cannot be read back and the counter shows 2 blocks.

The "sorry: constant selects" line is a known Icarus Verilog limitation message about the always_comb block in aes_axi_wrapper.sv. It does not change the result. The four "VCD warning: array word" lines only say that the waveform file names the result registers out_words[0] to [3] with escaped names; they come from wave_dump.sv, not from the design.

Transcript

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Experiment 3 - Pipelined AES-256 Encryption Engine (PYNQ-Z2)
Simulation transcript, 2026-09-18 15:50
Simulator: Icarus Verilog version 12.0 (devel) (s20150603-1539-g2693dd32b)

> iverilog -g2012 -o aes_sim.vvp -s aes_tb -s wave_dump ../RTL/*.sv ../Testbench/aes_tb.sv wave_dump.sv   (r
un inside Simulation/)
../RTL/aes_axi_wrapper.sv:164: sorry: constant selects in always_* processes are not currently supported (al
l bits will be included).

> vvp aes_sim.vvp
VCD info: dumpfile aes_wave.vcd opened for output.
VCD warning: array word aes_tb.dut.u_axi_wrapper.out_words[0] will conflict with an escaped identifier.
VCD warning: array word aes_tb.dut.u_axi_wrapper.out_words[1] will conflict with an escaped identifier.
VCD warning: array word aes_tb.dut.u_axi_wrapper.out_words[2] will conflict with an escaped identifier.
VCD warning: array word aes_tb.dut.u_axi_wrapper.out_words[3] will conflict with an escaped identifier.
key schedule ready (status = 00000004)
plaintext  00112233445566778899aabbccddeeff
ciphertext 8ea2b7ca516745bfeafc49904b496089
[PASS] encryption matches the FIPS-197 test vector
decrypted  00112233445566778899aabbccddeeff
[PASS] decryption recovered the original block
[PASS] the key registers read back as zero
blocks processed: 2
ALL AES-256 VERIFICATION TESTS PASSED SUCCESSFULLY
../Testbench/aes_tb.sv:151: $finish called at 2305000 (1ps)
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4. Waveform

(a) The last key words are written and CTRL = 4 pulses *ctrl_load_key*. The key expansion state machine goes to EXPAND and builds round keys 2 to 14, one per clock (*round_idx* 2 ... 14), then goes to READY; *key_ready* and LD0 come on. (b) Encryption: the block 00112233...eeff is XORed with round key 0 and enters round 1; each round is one pipeline register, so the round outputs change one clock after each other. Fourteen clocks after *start*, *r14 valid* pulses and *block_out* holds 8EA2B7CA...4B496089, which the AXI wrapper copies into the result registers; *busy* is high while the block is in the pipeline and the block counter goes to 1. (c) Software reads the four result words. (d) The same pipeline in decrypt mode turns the ciphertext back into 00112233...EEFF.

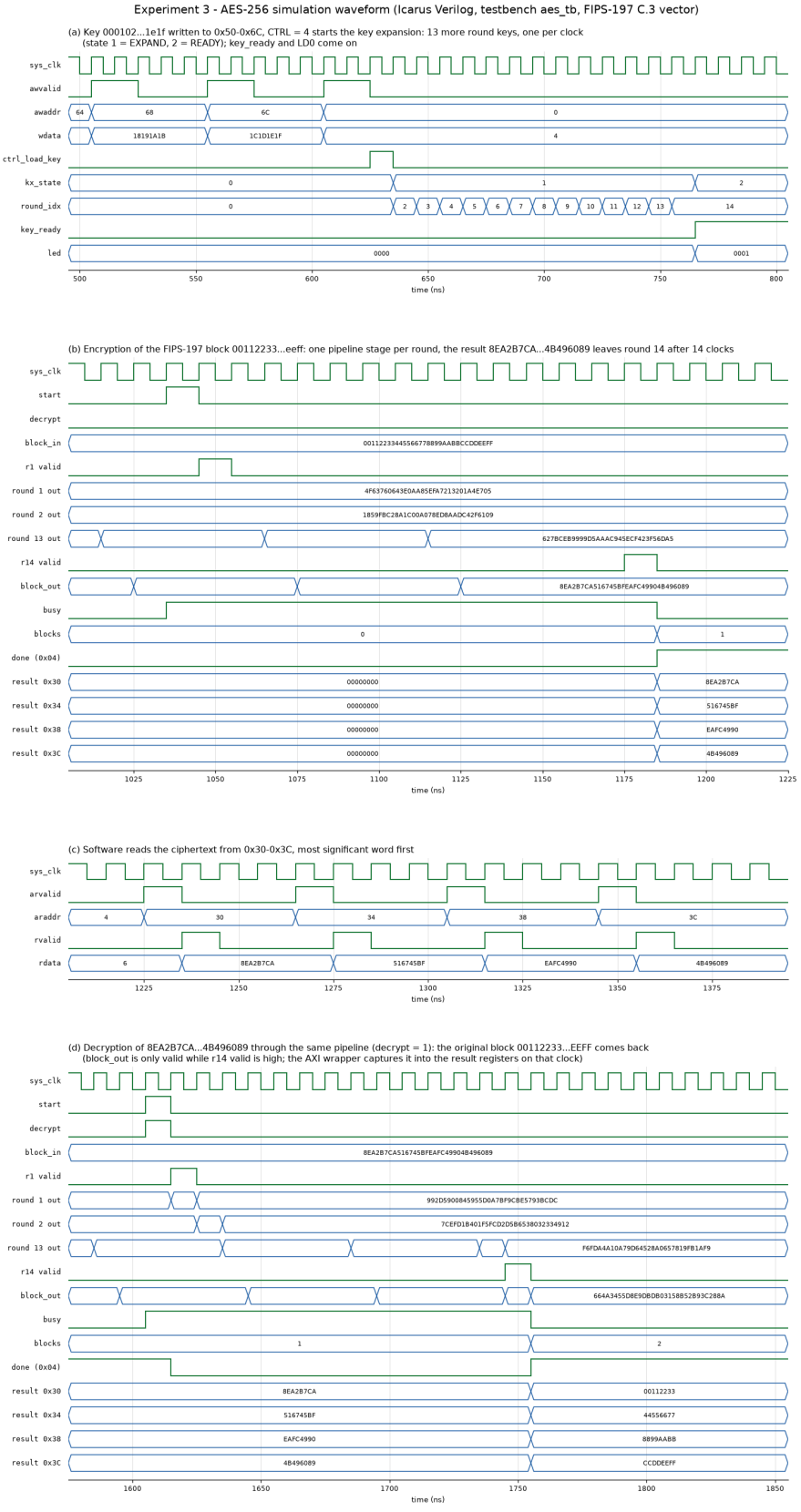


Figure 1 - Simulation waveform of aes_tb (drawn from aes_wave.vcd)